Mustafa Alshehab

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github.com/MustafaAlshehab

Senior Software Engineer | Automation | MCUs | DevOps | Cloud | AI

Senior software engineer with 8 years of experience specializing in end-to-end automation, firmware validation, modernizing codebases, improving reliability, and collaborating across engineering, QA, and stakeholders through production support.

AREAS OF EXPERTISE

Languages: C/C++, Python | **App Dev:** Qt/QML (desktop), FastAPI, REST APIs

Cloud/DevOps: AWS, Docker, CI/CD (Jenkins, Git/GitHub), Automation, Kubernetes, Ansible, Bash, PowerShell

Data/Build: PostgreSQL, SQLite, SQL | CMake, Packaging/Distribution, Windows build/test

Reliability/Security: production support, availability ownership, access controls | **OS:** Windows, Linux, macOS

GenAI tools: Cursor, Copilot, RAG | **Quality:** test automation, performance analysis, debugging

PROFESSIONAL EXPERIENCE

Washington State Board for Community and Technical Colleges | **Olympia, WA** | **April 2025 - Present**

Lead design, development, and maintenance of the Open Prison Education (OPE) platform, spanning automation tools, applications (web and desktops), and services supporting education across 10 Washington State correctional facilities and 5,000+ students per year in the state of Washington.

- Developed a Python CLI credentials tool that automated student-laptop provisioning via JSON-based configuration with early input validation/error handling, reducing setup time by ~75%; packaged and shipped as a standalone Windows executable.
- Led the OPE Server deployment overhaul—interactive YAML-driven setup, end-to-end documentation, and repository organization/cleanup—making deployments easier and more efficient by reducing manual, error-prone steps.
- Modernized the codebase and repo structure by splitting a monolithic repository into three modular components and rewriting git history, reducing repository size by 99% and accelerating onboarding and day-to-day development.
- Designed and implemented AWS EC2 infrastructure for OPE: Linux EC2 for prod/dev servers and Windows EC2 for RDP-based build/test environments, with IP-allowlisted SSH/RDP, SSM, and daily security patching.

Intel | **Hillsboro, OR** | **December 2021 - November 2024**

Led development and maintenance of automation for Windows driver certification of Intel Wi-Fi/Bluetooth connectivity chipset on laptops, partnering with cross-functional teams to ensure compliance and drive issue triage, and reporting. In addition, Improved workload integration, optimization, and benchmarking across Intel Xeon and competing platforms to demonstrate platform capabilities and strengthen customer confidence.

- Led the development of a new solution to fully automate the interaction with Microsoft Partners API, resulting in up to ~75% increased efficiency and fewer errors when distributing submission to OEMs.
- Managed multiple nightly Wi-Fi/Bluetooth testbeds, maintaining 99.9% uptime for internal clients during critical product launches.
- Developed and maintained comprehensive workload performance collaterals, ensuring accuracy and timely delivery, to showcase platform features, resulting in increased customer interaction and adaptation.
- Improved Wazuh workload execution efficiency by 73%, resulting in a substantial reduction of benchmarking time, validation processes, operational cost, and increased availability of resources.
- Identified and reported a security risk in the git history of an external software release, helping safeguard client integrity and prevent costly downstream impact.

- Led a successful microservices hackathon, fostering collaboration and reducing redundancy among internal teams by promoting early exposure to microservices and enhancing our internal catalog.
- Started and finalized “FAQ Revamped” initiative for the Workload Service Framework project, improving information discoverability and team productivity.

onsemi **Beaverton, OR** **June 2018 – November 2021**

Led the Hello Strata project end-to-end, accelerating Strata Developer Studio environment setup, deployment, and firmware development workflows to improve developer productivity and user experience. Enabled platform support to include bare-metal microcontroller (MCU) targets and contributed across the full development lifecycle—from feature development and validation to release. Developed and automated comprehensive test plans for hardware and applications to ensure product quality.

- Developed a feature for Strata Developer Studio to communicate with a client wirelessly within the local network, an addition to the existing wired USB connection. Resulting in enhanced user experience and increased customer adoption.
- Reduced start of Strata development time by 90% by making an SDK installer with concise documentation.
- Reduced building time of Strata Developer Studio by 70% after the initial build, significantly improving development productivity time and reducing operational costs.
- Automated deployment end-to-end using Python, Jenkins, Bitbucket Pipelines, and PowerShell, reducing deployment time by 60%.
- Reduced dependencies conflict by 50% by eliminating multiple OS build support utilizing Docker containers.
- Developed a firmware for bare-metal MCUs enabling serial communication with Strata Developer Studio increasing the number of Strata enabled MCUs.
- Developed React components to show graphical analytics data to be used in the main analytics page.
- Automated firmware build process by using Bash scripts, CMake, and a modified VS Code user interface.
- Created and maintained Linux Ubuntu Docker image with the tools/dependencies to build firmware.

Oregon State University **Corvallis, OR** **September 2016 - March 2017**
Teaching Assistant

- Supported course delivery by running weekly student help sessions and providing guidance on assignments.
- Evaluated student work and delivered detailed, constructive feedback to support learning outcomes.

Personal Projects

RAG Chatbot (rag_chatbot_addy_blog)

- Developed a full stack Retrieval Augmented Generation (RAG) application to facilitate semantic querying against external documentation, demonstrating advanced understanding of knowledge retrieval pipelines using tools and frameworks like **LangChaig**, **ChromaD**, **Gemma 4**, **Streamlit**.

Portfolio Website (MustafaAlshehab.github.io)

- Implemented a fully automated personal portfolio site deployed via GitHub Pages, demonstrating foundational understanding of continuous deployment and **GitHub Actions, Hugo, Yaml, Git**

EDUCATION

Bachelor of Science, BS, in Electrical and Computer Engineering, Oregon State University, Corvallis, OR